

Execution-Conditioned Reference Handoff for Asynchronous Humanoid Vision-and-Language Navigation

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Abstract—High-level planning and low-level locomotion control in humanoid vision-and-language navigation typically operate at different time scales. Asynchronous execution allows the robot to continue moving while the planner generates a new reference trajectory, but changes in body state and execution progress can make a geometrically feasible reference unsuitable for immediate execution upon arrival. To address this issue, we propose an execution-aware reference handoff framework (ECH) that explicitly decouples reference arrival from execution acceptance while preserving the existing navigation and locomotion stack. The framework comprises Executable Reference Adaptation (ERA) and Execution-Conditioned Reference Handoff (ECRH). ERA prepares the pending reference by adapting its local geometry to humanoid motion constraints while preserving the original navigation intent. ECRH determines when and how to adopt it by comparing the short-horizon execution outcomes of KEEP, UPDATE, and TRANSITION under safety constraints, conditioned on the robot’s current execution state. Within ECRH, the Executable Transition Mechanism (ETM) constructs a local transition trajectory connecting the ongoing motion to the adapted reference, providing the TRANSITION candidate when direct entry is unsuitable. Indoor stair-navigation experiments with a Unitree G1 in simulation show that, in challenging scenarios involving corners, stair ascent, and transitions from stair ascent to descent, the proposed approach avoids directly adopting new navigation references in unsuitable motion states. This enables more appropriate acceptance and execution of navigation intent, thereby reducing collisions, loss of balance, and falls.

I. INTRODUCTION

Vision-and-language navigation (VLN) aims to translate visual observations and language instructions into robot navigation behaviors. Continuous-environment benchmarks extend VLN beyond navigation graphs [1], while video-based and streaming approaches exploit visual history for navigation [2], [3]. Recent hierarchical navigation systems further integrate high-level semantic reasoning with local motion generation. NaVILA links vision-language reasoning to robot motion commands [4], while DualVLN coordinates a low-frequency semantic planner with a high-frequency local trajectory generator to support asynchronous reference updates during continuous execution [5]. For humanoid robots, newly generated navigation references must also be compatible with the robot’s continuously evolving physical state.

Existing studies address different aspects of this execution problem. Real-time action chunking allows the robot to continue executing committed actions while new actions are

being generated, reducing execution interruptions caused by asynchronous inference [6]. VLASH further accounts for the future state at which an action will actually be executed, mitigating the state mismatch between planning and execution [7]. From the embodiment perspective, VAMOS incorporates robot capabilities into navigation reasoning [8], while VLN-PE studies the effects of dynamics and control errors on physical navigation execution [9]. HumanoidVLN extends vision-language navigation to physically simulated humanoid robots [10], and Hiking demonstrates robust whole-body locomotion over challenging terrain [11]. At the control level, reference governors regulate reference inputs to satisfy system constraints [12], predictive safety certification assesses candidate controls through short-horizon prediction [13], and shielding filters unsafe actions before execution [14]. These mechanisms fit a broader safety-filter perspective [15], while Neural Simplex explicitly switches between neural and safety-oriented controllers [16]. Complementary consequence-based methods use imagined rollouts to guide decisions [17], predict action-conditioned event cues for navigation [18], or refine actions using latent predicted futures [19].

Despite these advances, a key problem in asynchronous navigation remains insufficiently addressed: when a new navigation reference arrives asynchronously while the humanoid is already moving, should it be adopted immediately, deferred, or connected through a local transition? Even a geometrically feasible reference may be unsuitable for direct entry from the robot’s current body state and execution progress.

Consider a robot following active reference A while the navigation module generates pending reference B from an earlier observation. During inference, the robot continues along A , so its position, motion state, and execution progress change before B arrives. Arrival alone does not determine how the ongoing motion should enter B : if direct entry is unsuitable, the robot continues along A until conditions improve or connects to B through a local transition. Direct updating is appropriate only when the current state permits it. We formulate this decision as an execution-conditioned reference handoff problem.

We propose an execution-aware reference handoff layer between navigation and locomotion that decouples asynchronous reference arrival from acceptance for physical execution. **Executable Reference Adaptation (ERA)** adjusts pending-reference geometry to humanoid motion constraints. When direct entry is unsuitable, the **Executable Transition Mechanism (ETM)** constructs a local connector. **Execution-**

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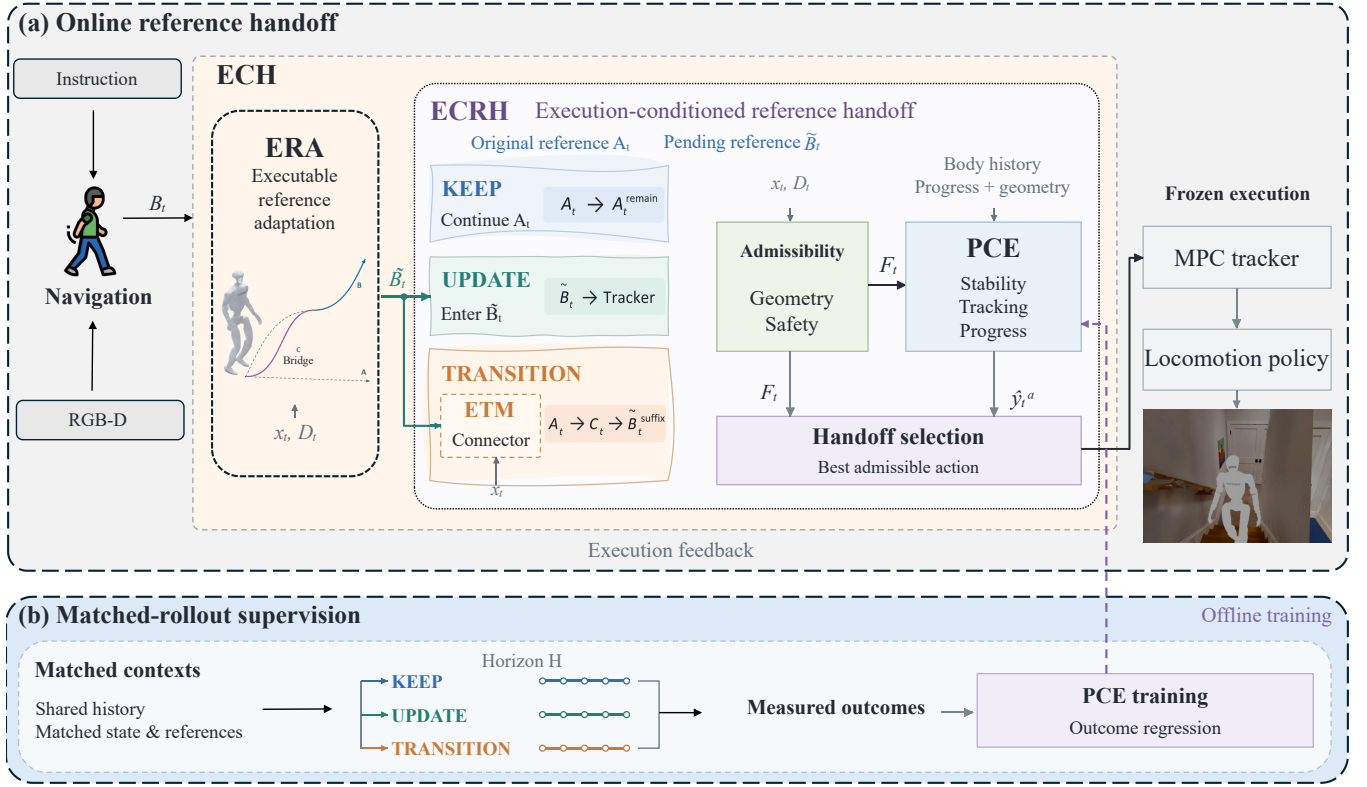


Fig. 1. Execution-aware reference handoff framework. (a) ERA adapts the pending reference, while ECRH selects among KEEP, UPDATE, and TRANSITION based on admissibility and PCE-predicted short-horizon outcomes. ETM constructs the connector for TRANSITION, and the selected reference is executed by the frozen tracker and locomotion policy. (b) PCE is trained offline using matched rollouts of the three handoff behaviors under shared states and references.

Conditioned Reference Handoff (ECRH) compares the short-horizon physical consequences of **KEEP**, **UPDATE**, and **TRANSITION** under independent admissibility constraints. Matched physical rollouts provide measured stability, tracking quality, and task progress to train the Physical Consequence Evaluator (PCE). Controlled and indoor navigation experiments compare the complete framework with direct reference installation during complex motion.

Our main contributions are:

- We formulate execution-conditioned reference handoff for asynchronous humanoid VLN, decoupling reference arrival from execution acceptance and modeling handoff as KEEP, UPDATE, or TRANSITION.
- We propose an **Execution-Aware Handoff (ECH)** framework, consisting of ERA for reference adaptation and ECRH for handoff decision. Within ECRH, ETM realizes TRANSITION by constructing an executable local transition reference.
- We introduce matched-rollout physical consequence supervision to train PCE from measured motion stability, tracking quality, and task progress.

II. METHOD

A. Overall Framework and Problem Formulation

At each handoff, the system considers active reference A_t , pending reference B_t , and the robot’s proprioceptive state, execution progress, and depth observations to determine

whether the new reference suits the current motion state. We select among three behaviors:

$$a_t \in \{\text{KEEP}, \text{UPDATE}, \text{TRANSITION}\}. \quad (1)$$

KEEP continues the active suffix while retaining the pending request; UPDATE directly adopts the adapted reference; and TRANSITION enters it through a local connector.

The complete method, ECH (Fig. 1), constructs candidate execution references, filters them through geometric and hard-safety checks, predicts their short-horizon consequences, and sends the selected reference to the existing tracker. The layer manages reference acceptance and switching while preserving the high-level navigation goal and underlying locomotion policy.

We distinguish four stages: *reference arrival* adds B_t to the candidate set; *execution acceptance* grants authority to a candidate; *reference installation* passes it to the tracker; and *physical convergence* occurs as the robot subsequently tracks and enters it. These stages need not coincide in time.

B. Execution-Aware Reference Handoff

Because the pending reference B_t is generated from an earlier observation while the robot continues moving during asynchronous inference, its local geometry may no longer align with the current execution state. Executable Reference Adaptation (ERA) first aligns B_t with the current execution context and removes obsolete segments. It then adapts local

heading, curvature, backward motion, and infeasible segments according to humanoid motion constraints and environmental clearance, while preserving the original navigation intent. The adapted reference is

$$\tilde{B}_t = \text{ERA}(B_t, x_t, D_t), \quad (2)$$

where x_t denotes the current execution state and D_t provides local geometry and clearance information from depth observations. ERA only prepares the pending reference for subsequent handoff decisions; it does not determine whether that reference should immediately acquire execution authority.

In addition to UPDATE, which directly adopts $\tilde{B}_t$, the Executable Transition Mechanism (ETM) constructs a local connector C_t for TRANSITION. Starting from the current motion state, this connector gradually joins a feasible entry location on $\tilde{B}_t$, yielding

$$R_t^T = C_t \oplus \tilde{B}_t^{\text{suffix}}, \quad (3)$$

where $\oplus$ denotes trajectory concatenation and $\tilde{B}_t^{\text{suffix}}$ is the executable suffix of the pending reference beyond the entry location. ETM preserves the ongoing direction of motion where possible and gradually aligns it with the local direction of the new reference, reducing the geometric discontinuity caused by direct switching. A successfully constructed TRANSITION candidate undergoes admissibility checks alongside KEEP and UPDATE.

Once the candidate execution references are prepared, Execution-Conditioned Reference Handoff (ECRH) makes a unified decision among the three behaviors. Independent geometric and hard-safety checks use depth observations and whole-body state to exclude currently inadmissible candidates, yielding the admissible behavior set $\mathcal{F}_t$. Within ECRH, the Physical Consequence Evaluator (PCE) then uses recent proprioceptive state, execution progress, and local candidate-reference geometry to predict the short-horizon execution outcomes of each admissible behavior:

$$\hat{\mathbf{y}}_t^a = [\hat{y}_{\text{stab}}^a, \hat{y}_{\text{track}}^a, \hat{y}_{\text{prog}}^a], \quad (4)$$

where the three terms represent motion stability, trajectory tracking quality, and task progress, respectively. PCE does not directly use raw RGB or depth images as learned inputs; it evaluates candidate consequences from execution state and reference geometry. ECRH then selects the final behavior from the admissible set:

$$a_t^* = \arg \max_{a \in \mathcal{F}_t} S(\hat{\mathbf{y}}_t^a), \quad (5)$$

where $S(\cdot)$ jointly evaluates motion stability, trajectory tracking quality, and task progress.

Together, ERA and ETM prepare candidate references, while ECRH applies admissibility filtering, consequence prediction, and behavior selection to determine when and how the pending reference is entered.

C. Consequence Supervision from Matched Physical Rollouts

We train PCE using matched physical rollouts of KEEP, UPDATE, and TRANSITION over a fixed horizon. Rollouts share a common history prefix, with pre-handoff robot states and references matched as closely as possible to compare execution outcomes. Exact restoration of the simulator's complete internal state is not required.

For a candidate behavior a , the supervision target is

$$\mathbf{y}_t^a = [y_{\text{stab}}^a, y_{\text{track}}^a, y_{\text{prog}}^a], \quad (6)$$

corresponding to motion stability, trajectory tracking quality, and task progress measured during the rollout. PCE learns these short-horizon execution outcomes through multi-target regression:

$$\mathcal{L}_{\text{PCE}} = \lambda_s \mathcal{L}_{\text{stab}} + \lambda_t \mathcal{L}_{\text{track}} + \lambda_p \mathcal{L}_{\text{prog}}, \quad (7)$$

where λ_s , λ_t , and λ_p balance the supervision weights of the different outcome targets.

This supervision directly teaches the short-horizon execution outcomes of candidate handoff behaviors rather than the final action class. No additional physical rollouts are required during online execution: the trained PCE directly predicts outcomes for admissible candidates, and ECRH makes the final handoff decision.

III. EXPERIMENTS

A. Experimental Setup

We evaluate the proposed reference-handoff method on a Unitree G1 in NVIDIA Isaac simulation. Indoor navigation examples use Matterport3D-derived assets [20]. The navigation, trajectory-tracking, and locomotion components are kept identical across handoff comparisons; only the mechanism by which a newly generated reference acquires execution authority is changed.

We use **Direct Update** as the primary baseline, replacing the active reference immediately upon the arrival of a new reference. In contrast, our method mediates reference entry through KEEP, UPDATE, and TRANSITION.

We consider three controlled handoff scenarios: a 90° turn, a narrow corridor, and uneven terrain. Each method is evaluated over ten trials per scenario. We report **success rate**, **final goal error**, and **handoff delay**. Handoff delay is defined as the interval from the arrival of reference B to its installation in the trajectory tracker, excluding the subsequent time required for physical convergence to the reference trajectory.

B. Controlled Handoff Results

Table I summarizes the controlled comparisons. Direct Update succeeds in 5/10 trials for the 90° turn, 3/10 in the corridor, and 2/10 near the raised edge, whereas ECH succeeds in all ten trials in each setting. Final goal error is reduced from 2.68 to 0.18 m, 2.25 to 0.33 m, and 2.90 to 0.25 m, respectively.

These improvements introduce average handoff delays of 1.15 s, 3.83 s, and 4.94 s across the three scenarios.

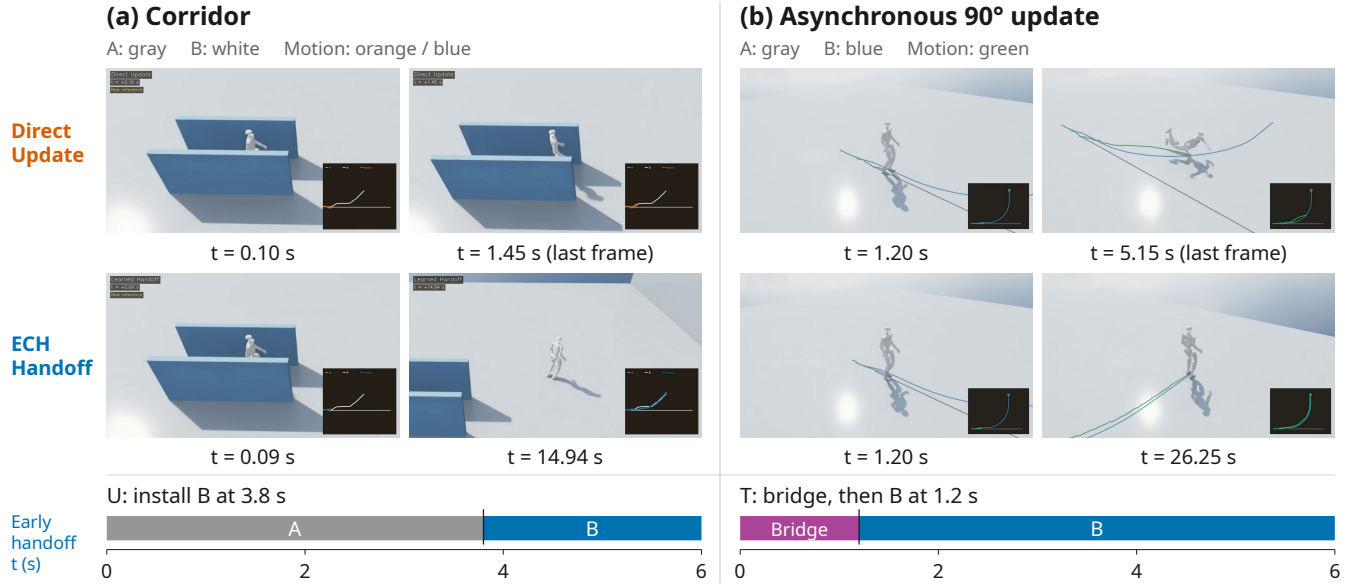


Fig. 2. Representative handoff behaviors. (a) In the corridor, ECH retains the active reference until the pending route becomes suitable for entry. (b) For a large heading change, ECH uses an explicit transition reference to connect the ongoing motion to the pending route. Timestamps are relative to reference arrival.

TABLE I

CONTROLLED HANDOFF RESULTS OVER 10 TRIALS PER METHOD AND SCENARIO. SUCCESS RATES ARE REPORTED AS PROPORTIONS. HANDOFF DELAY IS REPORTED ONLY FOR ECH.

Scenario	Method	Success rate $\uparrow$	Goal error (m) $\downarrow$	Handoff delay (s)
90° turn	Direct	0.50	2.68 ± 2.52	–
	ECH	1.00	0.18 ± 0.07	1.15 ± 0.20
Corridor	Direct	0.30	2.25 ± 1.34	–
	ECH	1.00	0.33 ± 0.10	3.83 ± 0.56
Raised-edge	Direct	0.20	2.90 ± 1.39	–
	ECH	1.00	0.25 ± 0.13	4.94 ± 0.71

Rather than indicating rejection of the new reference, the delay reflects deferred entry when the pending reference is temporarily unsuitable for the current execution state, while its navigation intent is preserved.

In the narrow corridor (Fig. 2(a)), Direct Update changes the motion direction while the robot is still constrained by the corridor geometry, causing it to approach the wall. In contrast, ECH maintains KEEP and follows the active reference until a more suitable entry state is reached, after which the pending reference is adopted. This preserves the new navigation intent while deferring its physical execution.

For a large heading change (Fig. 2(b)), direct replacement introduces an abrupt execution disturbance that can destabilize the robot. ECH instead selects TRANSITION, with ETM constructing a local connector that gradually links the ongoing motion to the pending reference. This allows the handoff to regulate not only *when* the new reference is entered, but also *how*.

Near the raised edge (Fig. 3), Direct Update causes contact with the terrain edge, followed by stumbling or loss of stability. In contrast, ECH retains the ongoing motion before entering the pending reference through a transition.

TABLE II

REPRESENTATIVE RAISED-EDGE EXECUTION. GOAL-DISTANCE REDUCTION IS MEASURED AT THE RECORDED ENDPOINT OF EACH RUN.

Metric	Direct Update	ECH
Handoff delay (s)	0.00	4.90
Goal-distance reduction (%)	14.6	97.5
Execution outcome	Terminated	Successful

In the representative run in Table II, ECH achieves substantially greater goal-distance reduction and completes execution, whereas Direct Update terminates early. This further shows that deferring and restructuring reference entry can improve execution when local terrain restricts immediate motion changes.

C. Indoor Navigation Examples

We further examine the proposed method in longer indoor sequences involving stairs, corners, and changes in navigation intent.

A separate locomotion-backend comparison (Fig. 4(a)) shows that Hiking provides the terrain-traversal capability required by the subsequent handoff experiments. This comparison characterizes the underlying locomotion capability and is not treated as evidence of performance gains from the reference-handoff method itself.

At a tight corner (Fig. 4(b)), directly adopting the new reference causes the robot to turn before sufficient space is available, resulting in wall contact and blocking further motion. In contrast, execution-aware reference handoff allows the robot to complete the turn and continue navigation.

During stair ascent (Fig. 4(c)), Direct Update causes repeated contact with the railing and eventually leads to downward slipping. With ECH, the robot continues to the landing, completes the turn there, and proceeds toward the

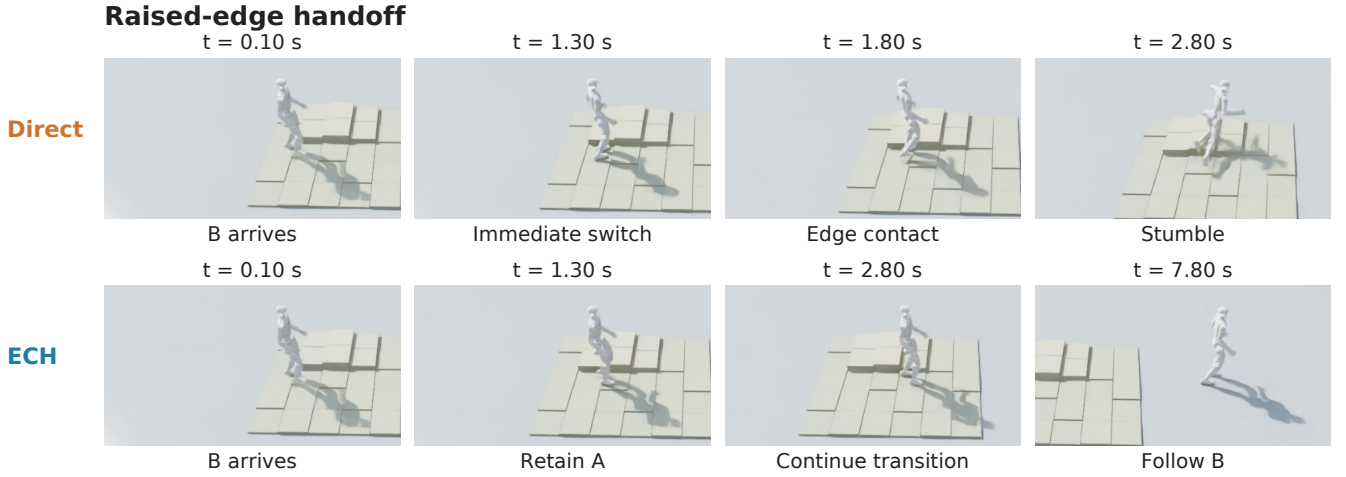


Fig. 3. Reference handoff near a raised edge. Direct replacement leads to edge contact and loss of stability, whereas ECH retains the current motion and enters the pending route through a transition.

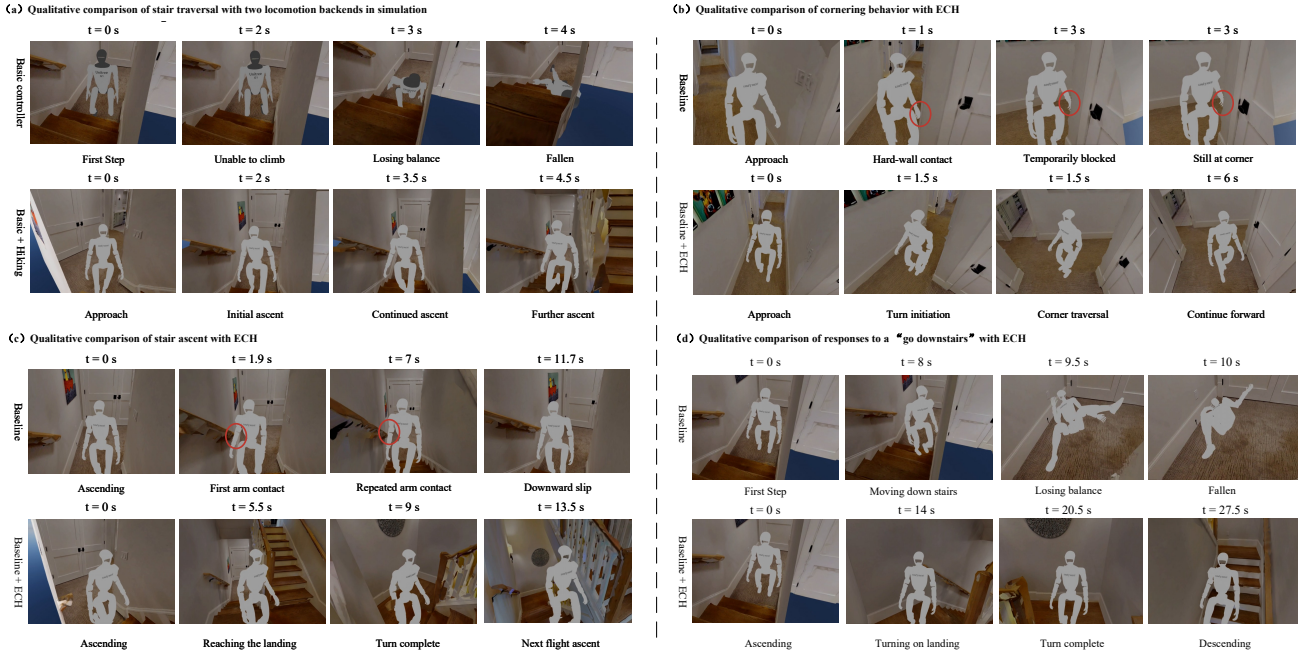


Fig. 4. Indoor simulation examples. (a) Locomotion-backend comparison. (b) Corner traversal. (c) Stair ascent with railing interaction. (d) A new descent request received during ascent. Panels (b)–(d) compare direct execution with execution-aware handoff.

next flight of stairs.

A more representative case occurs when a descent instruction is received during stair ascent (Fig. 4(d)). Direct Update introduces the descent reference while the robot is still on the stairs, causing premature reversal without sufficient turning space and ultimately leading to loss of balance. In contrast, ECH retains the new descent intent while continuing the ongoing ascent until the robot reaches the landing. It then turns in the available space and enters the descent reference.

These examples illustrate that reference handoff is not simply an added waiting period. Instead, it determines *when* and *how* a new reference should enter ongoing execution according to the current motion state, while preserving the navigation intent.

IV. DISCUSSION

The experimental results indicate that reference handoff does more than simply delay the execution of a new reference: it coordinates reference changes during continuous motion. In a narrow corridor, continuing along the active reference *A* maintains progress until sufficient turning space becomes available. For a large heading change, an explicit transition trajectory provides a more suitable connection to the new reference *B*. On uneven terrain or near a raised edge, deferring direct switching avoids abrupt changes while the robot remains subject to local physical constraints. These observations support our central premise: a geometrically valid reference trajectory may still be unsuitable for immediate execution. Handoff delay should therefore be interpreted to-

gether with the motion actually executed during that interval. Installing a reference in the tracker and physically converging to that reference are distinct events; an earlier switch does not necessarily imply earlier or more reliable completion.

The indoor stair experiments further illustrate the distinction between navigation intent and physical execution. A new navigation request can be retained while its execution is deferred until the robot reaches a more suitable motion state, such as a stair landing with sufficient turning space. ECRH compares admissible handoff behaviors using predicted short-horizon stability, trajectory tracking quality, and task progress, while geometric constraints and hard-safety checks remain independent of the learned consequence evaluator. The current experiments are primarily conducted with a Unitree G1 in simulation and demonstrate the framework's effectiveness under representative challenging motion conditions involving turns, stairs, and terrain changes. Future work will investigate the compatibility of the handoff layer with different high-level navigation modules and underlying locomotion controllers, extend evaluation to a broader range of terrains, navigation tasks, and motion-state distributions, and pursue deployment and validation on physical humanoid platforms.

V. CONCLUSION

We presented an execution-aware reference handoff framework for asynchronous humanoid navigation that explicitly decouples asynchronous reference arrival from acceptance for execution. ERA adapts the pending reference to the current execution context, ETM constructs an executable local transition reference when direct entry is unsuitable, and ECRH compares the short-horizon physical consequences of alternative handoff behaviors under safety constraints to determine when and how the new reference should be entered. Controlled experiments and indoor simulation results show that, compared with direct reference replacement, the proposed method better coordinates the integration of new references into ongoing motion, improving execution stability and robustness under challenging motion conditions. These findings highlight reference handoff as an important interface between asynchronous navigation planning and humanoid physical execution.

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